Automation in Construction 132 (2021) 103961

Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon

Digital twin-driven virtual sensor approach for safe construction operations of trailing suction hopper dredger

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ARTICLE INFO

Keywords:
Trailing suction hopper dredger
Digital twin
Virtual sensor
Machine learning
Data mining
Construction safety

ABSTRACT

The stable and safe operation of Trailing suction hopper dredger (TSHD) is one of the most crucial considerations for ensuring its high dredging productivity. However, the instability and sudden failure of physical sensors pose challenges to the monitoring of dredging process. To address these issues, we propose a structure of digital twin-driven virtual sensor (DTDVS) for the construction safety of TSHD. Considering the potential internal relations among construction data, we compare the performance of four machine learning algorithms in predicting the torsional vibration in mechanical failure. The results showed that these algorithms provide high prediction accuracy ($R^2 > 0.9$). Then the DBN model with the best performance was selected as a part of the virtual sensors to predict and analyze the status of TSHD. The digital twin technology provides a more stable and environmentally friendly scheme for TSHD construction safety control. On the one hand, the DTDVS assists physical sensors to monitor the construction state, overcoming the limitation of the sensors on detection targets which are difficult or costly to measure directly. On the other hand, by analyzing the residual between the physical sensor and the virtual sensor, the construction behavior can be diagnosed, and the fault situation can be pre-warned accurately. This improves the time utilization of TSHD and provides an important guarantee for the construction safety.

1. Introduction

Dredging engineering involves excavating the soil of the river or seabed and then transport the excavated soil to a designated dumping area [1]. A piece of important dredging equipment, trailing suction hopper dredger, is often operated in the sea with complex soil and a harsh environment [2]. Therefore, it is challenging to accurately characterize the TSHD behavior. During each construction period, the random load caused by gradual loading, uncertain soil quality, the wind and wave [3] may affect the mechanical properties, reduce the dredging efficiency and endanger the construction safety. Therefore, real-time monitoring, timely diagnosing and effective pre-warning of the construction status are essential for the efficient operation of the TSHD.

Physical sensors are important means to monitor the construction status. Damen [4] used a series of high-precision measuring instruments such as flowmeter and nuclear densimeter, to detect the construction status of dredgers and further obtained a large amount of real-time monitoring data. Park et al. [5] proposed a cloud-based real-time security monitoring system framework to collect and analyze field conditions data and communicate through the cloud. Based on an optimal flow of real-time information, Skibniewski et al. [6] proposed a project management framework for dredging projects to enhances decision making and overall control of project performance. Although the physical sensors play an important role in construction safety control of the TSHD, they have fundamental limitations: (1) the physical sensors are at risk of instability and failure caused by complex construction environment; (2) establishing the simulation model solely based on the physical measurements constrains determining the relationship between physical quantities; (3) there are still several construction parameters that are too difficult or expensive to be directly measured, e.g., the real-time mud concentration in the trailing suction pipe. Furthermore, the traditional measurement techniques have a risk of nuclear radiation and environmental pollution due to the use of radioactive sources [7]. Therefore, it is necessary to propose a stable and reliable new sensor to address the above problems.

The digital twin is the digital representation of the physical model [8], which realizes the reconstruction, deduction and analysis of the physical model in the digital scene [9]. The data flow between the physical system and the network is bidirectional and synchronous [10]. Tuegel et al. [11] proposed the concept of Digital twin in 2011, where

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https://doi.org/10.1016/j.autcon.2021.103961
Received 28 March 2021; Received in revised form 7 September 2021; Accepted 10 September 2021
Available online 21 September 2021
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Table 1
Monitoring data acquired by the physical sensor of TSHD.

Number	1	2	3	4	5	…	792	793
Name	Carrying capacity (t)	Drag depth (m)	Flow rate (m^3/h)	Density (t/m^3)	Rotational speed of engine (rpm)	…	Power of the pump (kW)	Stern draft (m)
0:00:01	21,226.88	13.86	21,461.67	1.11	597	…	2,030	10.87
0:00:03	21,225.68	13.88	21,476.52	1.11	606	…	1,897	10.87
0:00:05	21,221.11	13.84	21,461.67	1.11	597	…	2,197	10.88
0:00:07	21,218.11	13.79	21,469.09	1.11	598	…	2,399	10.87
0:00:09	21,214.37	13.68	21,456.72	1.12	600	…	2,450	10.88
0:00:11	21,212.74	13.75	21,471.57	1.12	600	…	2,418	10.88
0:00:13	21,210.68	13.75	21,449.29	1.13	599	…	2,447	10.88
0:00:15	21,210.25	13.80	21,449.29	1.13	598	…	2,451	10.88
0:00:17	21,211.60	13.70	21,451.77	1.13	599	…	2,424	10.88
0:00:19	21,209.68	13.55	21,454.25	1.13	600	…	2,404	10.88
0:00:21	21,207.76	13.52	21,459.19	1.13	599	…	2,408	10.87
0:00:23	21,210.31	13.51	21,459.19	1.13	598	…	2,442	10.87
0:00:25	21,213.32	13.53	21,449.29	1.13	599	…	2,404	10.88
0:00:27	21,215.63	13.48	21,449.29	1.13	599	…	2,405	10.88
0:00:29	21,217.79	13.47	21,461.67	1.13	599	…	2,368	10.88
…	…	…	…	…	…	…	…	…
23:59:57	21,419.78	13.17	20,793.47	1.12	601	…	2,237	11.18
23:59:59	21,417.04	13.16	20,786.04	1.10	597	…	2,239	11.22
Maximum	21,419.78	13.88	21,476.52	1.13	606	…	2,451	11.22
Minimum	21,207.76	13.16	20,786.04	1.10	597	…	1,897	10.87
Average	21,238.98	13.61	21,380.00	1.12	599	…	2,331	10.91

digital models were used to simulate and analyze aircraft behavior. Later, the digital twin was applied to aerospace [12–14] and manufacturing industries [15–17], and its application in construction safety was also explored. Wang et al. [18] proposed a digital twin reference model for fault diagnosis in rotating machinery. Then they proposed a model updating scheme based on parameter sensitivity analysis to further improve the adaptability of the model. Booyse et al. [19] directly extracted digital models from operating data by using deep learning, avoiding the limitation caused by deterministic physical models. Compared with the traditional regular maintenance inspection, the digital twin technology provides a new way for fault monitoring.

Deep learning plays a critical role in the digital twin. Hinton et al. [20] proposed the idea of deep learning in 2006. Compared with shallow neural networks, deep learning models directly use the original data as the input and then learn the features of the data layer by layer to achieve a more effective feature representation [21]. With the development of artificial intelligence, deep learning has been widely used in image processing [22], natural language processing [23], speech recognition [24], and other fields. In recent years, the use of deep learning has been examined for safety monitoring [25]. Bieman et al. [26] proposed a technique based on deep learning combined with video imagery of physical model tests to develop a non-intrusive measurement method. Furthermore, Bai et al. [27,28] proposed superimposing multiple models to improve the accuracy of productivity prediction for the TSHD. Tamilselvan et al. [29] and Tran et al. [30] also suggested a highly reliable health diagnosis method based on the DBN model which can be applied to the health diagnosis of aero-engines and power transformers. Huang et al. [31] combined the monitoring system with a forecasting model for operational monitoring and prediction of wave run-up on the seawalls. Tang et al. [32,33] also proposed a fault handling mechanism based on a rule base and a case base. The expert system designed identifies the cause of failures and then provides solutions. However, a rather large set of fault data and case support is required for their proposed approach. As seen in the above-mentioned researches, the existing research on fault detection is often based on the monitoring data of traditional physical sensors. In these techniques, the data flows in only one direction and feedback to the physical sensors does not exist. Therefore, the existing methods are unable to take full advantage of digital twin technology and machine learning. And the applicability of the above fault detection methods to the TSHD system needs to be studied further.

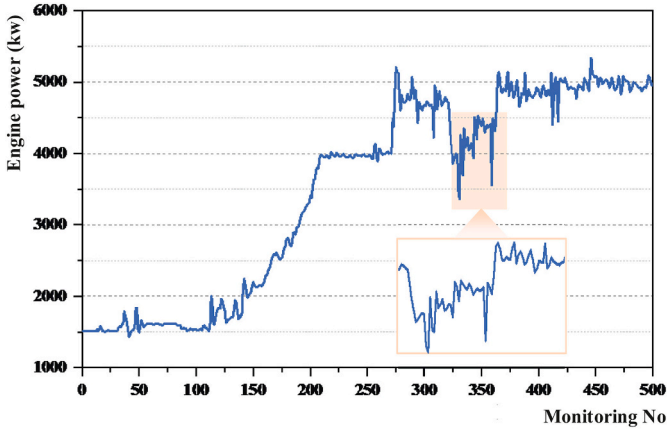


Fig. 1. Instances of the monitoring data of the main engine power.

In this paper, we assert that developing virtual sensing technology facilitates intelligent and efficient construction safety control of TSHD. Then by analyzing the real-time monitoring data, we propose a virtual sensor framework for TSHD based on the digital twin. In the virtual model, the operation indexes of mechanical equipment are predicted and evaluated. The virtual sensors assist physical sensors in accurate and timely diagnosis and further enables pre-warning of the potential mechanical faults.

2. General scheme of digital twin-driven virtual sensor

Virtual sensors are different from traditional physical sensors. They mainly aim at dynamic quantities which are difficult, expensive, or harmful to measure directly. Furthermore, in cases where the accuracy of physical sensors is declined or they become faulty in the construction process, the virtual sensors can replace the physical sensors for continuous monitoring of the construction status. Virtual signals are obtained through the internal relations among relevant construction indexes. The combination of virtual and physical sensors guarantees the stability of dynamic monitoring during construction. In addition, once a fault occurs in the construction process, the difference of parameters would lead to great differences between physical signals and virtual signals. Therefore, the residual value between physical sensors and virtual

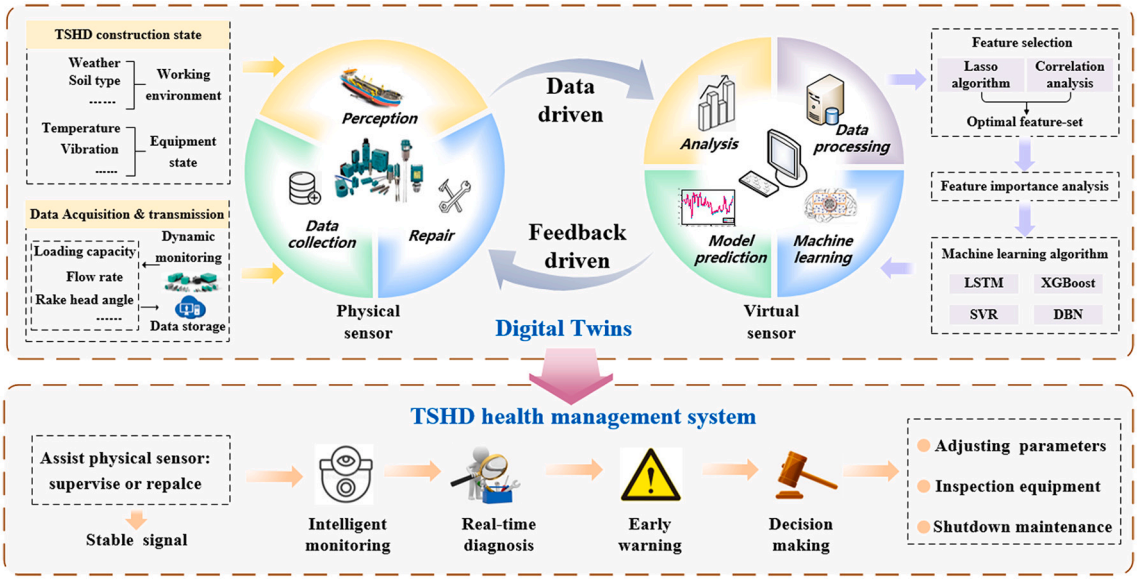


Fig. 2. The application of virtual sensors in construction safety control of TSHD based on digital twin.

sensors can be used to warn mechanical accidents and improve the safety of dredging operations.

2.1. Physical sensor

Modern large-scale trailing suction ships are often equipped with a supervisory control and data acquisition (SCADA) system. The basic life characteristic parameters of the ship, such as its loading capacity, main engine power, and rake head angle are obtained by physical sensors. The dredging operation status of the TSHD during the construction process is also fully inspected. Table 1 shows the construction monitoring data collected by the physical sensors, it is seen that the monitoring data contains a total of 793 dimensions. Furthermore, the data is time-dependent and noisy hence it may interfere with the data analysis, as shown in Fig. 1. Therefore, to reduce the influence of environmental factors and the instability of the instrument itself, it is necessary to preprocess the data before analysis to improve the data utilization efficiency.

The trailing suction dredger is equipped with a large number of physical sensors, which can directly reflect the dredging construction in real time. Although monitoring data can be obtained by physical sensors, if only the physical model is established as the main frame, it is difficult to determine the corresponding relationship between physical quantities, and the physical sensors have the risk of instability and failure. At the same time, some construction parameters are expensive or difficult to measure directly, such as the real-time mud concentration in the trailing suction pipe of the trailing suction dredger, the nuclear radiation risk and environmental pollution caused by the use of radioactive sources in the traditional measurement technology, and the risk of accidents and environmental damage in the measurement. Therefore, it is necessary to add the corresponding virtual sensors to assist the physical sensors through digital twin technology, so as to improve the accuracy of fault monitoring and reduce environmental pollution.

2.2. Overall framework

To overcome the shortcomings of physical sensors, the digital twin-driven virtual sensor approach is proposed. The digital twin includes three main components [34]: (a) physical model in real space; (b) digital model in virtual space; and (c) data and information connection that link the digital models with the physical systems. Here we propose the digital twin structure for heath management of TSHD, as illustrated in Fig. 2. The specific implementation steps are the following: (1) the physical sensors installed on dredgers dynamically sense and collect the TSHD construction status, severing as the foundation of digital model establishment and intelligent analysis; (2) the information management and control platform based on machine learning is the key to digital models. The virtual sensor preprocesses the construction data and makes a prediction based on the established model and further analyzes the results; (3) the virtual sensor provides feedback to the physical sensor. The real-time monitoring, diagnosis, and early warning analysis of the TSHD construction status are realized by integrating the two sensors.

2.3. Data preprocessing

2.3.1. Feature selection

In the feature selection, features that are closely related to the target are extracted from a large number of unknown relationship variables. The features are then used in building a model and improve the efficiency accuracy. The quality of the selected features depends on the correlation between the feature classes and the degree of feature redundancy. The higher the quality of the features, the more efficient the corresponding model.

As mentioned above, the construction monitoring data is high dimensional data, however, it is not clear which of the 793 monitoring data are the key parameters and which are the redundant parameters. Here we propose a method combining Lasso algorithm with random forest algorithm to select the best candidate parameters. Firstly, the Lasso algorithm is used to screen out the features that have a high correlation with the targets, including a large number of redundant features. Then, the redundant features are removed using the correlation heat map to obtain better features.

The basic idea of the Lasso algorithm is to minimize the sum of squares of residuals subject to the constraint on the sum of absolute values of the regression coefficients. This ensures generating a series of regression coefficients strictly equal to 0. The model consists of a linear model with a regularization term. Hence the minimized objective function is as the following [35]:

$$\min_{\omega} \frac{1}{2n}\|X\omega - y\|_2^2 + \alpha\|\omega\|_1 \tag{1}$$

where n is the number of samples, α is a constant value, $\|\omega\|_1$ and is a coefficient vector.

The LASSO constructs a penalty function to obtain a refined model,

which is beneficial in dealing with multicollinearity data. However, this method is unable to effectively deal with the redundancy of data, especially in the monitoring data of TSHD. If redundant features are used as the input features of the model, they will constitute redundancy and affect the speed and accuracy of machine learning. Therefore, to obtain the final preferred feature set, it is necessary to analyze the physical meaning of the features with higher correlation while removing the redundant features.

To further clarify the factors affecting the target, a random forest algorithm is used to analyze the importance of the features. The importance score measures the value of the feature in the decision tree model. The more the attributes are used to build a decision tree, the more important the features. Pareto chart can be also drawn according to the importance of the selected feature data. Features with a cumulative contribution rate of 0%–80% are the primary features. Besides, the features with a cumulative contribution rate of 80%–90% are called secondary features. The rest of the features are also called general features with a cumulative contribution rate of 90%–100%.

2.3.2. Savitzky-Golay filter

The TSHD is constructed in the complex marine environment and the sensor output may become unstable due to multiple factors such as waves, wind, and mechanical vibration. For instance, the measured drag head angle may contain cause noise due to the layout of monitoring equipment, surrounding temperature, sensitivity, and other reasons. Since these are often random noises, they can be smoothed using the Savitzky-Golay filter to improve the smoothness of data and reduce the noise interference.

Savitzky and Golay [36] proposed the Savitzky-Golay filter in 1964. It is a filtering method based on local polynomial least square fitting in the time domain to filter out the noise without changing the shape and width of the sequence. Considering 2n + 1 data points, the fitting polynomial is

$$P_m(i) = \sum_{k=0}^{m} b_k i^k \tag{2}$$

where m is the order of the fitting polynomial, b_k is the fitting coefficient, and k is the number of the fitting term. The residual sum of squares of the least-squares fitting is as the following:

$$SSE_m = \sum_{i=-n}^{n} \{P_m(i) - I(i)\}^2 \tag{3}$$

where n is the number of samples, $P_m(i)$ and $I(i)$ are the fitting and actual values, respectively.

2.4. Prediction and analysis

To ensure the harmony between the digital model and the physical world, it is essential to establish a digital twin model to match the physical system. Noting the potential internal relationship of the monitoring data of TSHD, here we combine the physical model with an artificial intelligent mode and propose the virtual sensor framework. The virtual sensor trains and predicts the detection target through machine learning to realize the auxiliary application of the physical sensors and to further improve the accuracy of the construction monitoring. Therefore, a highly accurate prediction model is critically important as it enables the application of virtual sensors.

2.4.1. Models of machine learning

Four machine learning algorithms with different properties are used to predict and analyze the data including the Long-Short Term Memory Network (LSTM), Extreme Gradient Boosting (XGBoost), Support Vector Regression (SVR), and Deep Belief Network (DBN). Besides, model evaluation indexes including R^2, root mean square error (*RMSE*), and mean absolute percentage error (*MAPE*), are adopted to evaluate and analyze the prediction performance of the models.

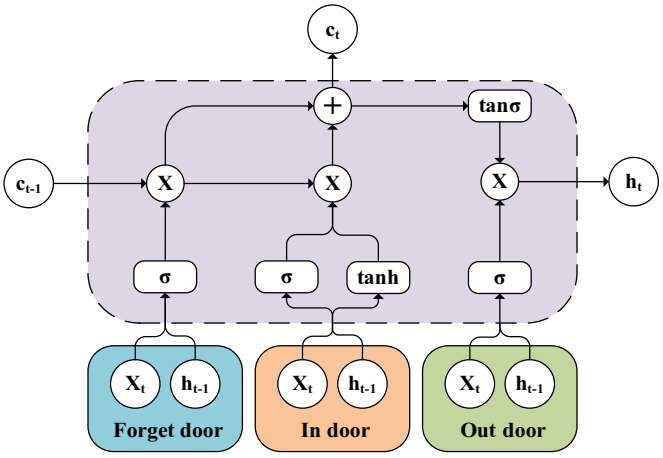


Fig. 3. The LSTM Structure.

2.4.1.1. LSTM. The LSTM is a special type of RNN, where the concept of time-series is incorporated into the design of the network structure to address the gradient disappearance and gradient explosion issues in the process of long sequence training [37]. The LSTM provides high robustness and generalization performance. Generally, the LSTM consists of three multiplication gates that control the retention and transmission of information. Fig. 3 illustrates the basic structure of the LSTM. By activating the input gate, the information of the new input data is being accumulated in the cell. The decision to whether or not to transfer c_t to the final state h_t also made in the output gate.

2.4.1.2. XGBoost. The XGBoost is an integrated learning method. The basic idea in XGBoost is to enhance the model's learning ability by iterative computation of multiple weak learners. The core idea is to constantly add trees and split the features to grow the tree. Adding one tree at a time is to learn a new function *f(x)* to fit the residual of the latest prediction. Finally, by adding up the scores corresponding to each tree, the predicted value of the sample is obtained. Here, the objective function is [38]:

$$L(\phi) = \sum_i l\left(y_i', y_i\right) + \sum_k \Omega(f_t) \tag{4}$$

where l is the loss function that reveals the training error, y_i' is the output of the accumulative model, y_i is the actual value and k represents the number of weak classifiers. In Eq. (4) $\sum_k \Omega(f_t)$ denotes the complexity function of the tree, for which the smaller the value, the lower the complexity and the stronger the generalization ability.

2.4.1.3. SVR. The SVR model utilizes the support vector machine (SVM) for curve fitting and regression analysis. Classification and regression are the two essential tasks in supervised machine learning. The SVM is highly efficient at solving small sample, nonlinear, and high-dimensional pattern recognition problems. The SVM classification obtains a plane so that the support vectors of the two classification sets or all the data are farthest away from the classification plane. The SVR regression is to find a regression plane, so that the distance from all the data of a set to the plane is the shortest [39].

2.4.1.4. DBN. The DBN is a classic deep learning network proposed by Hinton et al. in 2006, which is a tightly connected hierarchical structure formed by stacking multiple restricted Boltzmann machines (RBM) [40]. The RBM which is an unsupervised learning algorithm includes two-

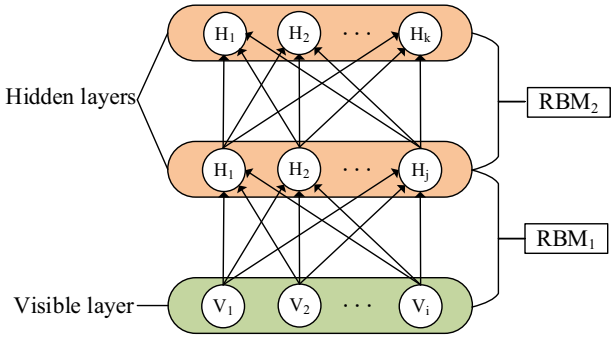


Fig. 4. The DBN Structure.

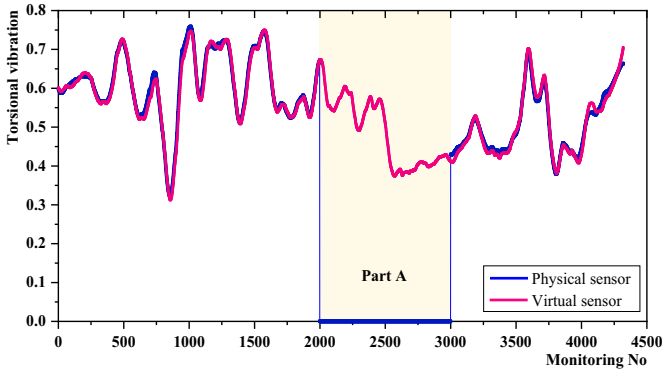


Fig. 5. Auxiliary applications of virtual sensors.

layer recurrent neural network that consists of a visible layer and a hidden layer where each layer contains a group of neurons or nodes. The DBN is trained in a greedy hierarchical manner by superimposing the RMBs together. The output of RBM_1 is used as the input to RBM_2 for training as shown in Fig. 4.

These machine learning algorithms provide an artificial intelligence-based method for data mapping. Choosing an appropriate model plays an important role in the efficient combination of the virtual and physical sensors in digital twin.

2.4.2. Model evaluation

For quantitative evaluation of the model's prediction performance, relevant indicators are used for evaluation and analysis. Error metrics are usually divided into longitudinal and lateral errors. The longitudinal error is to analyze the long-term operation of the system, e.g., *RMSE* and *MAPE*. The lateral error is used to characterize the overall prediction performance, e.g., Coefficient of determination (R^2). These indicators are defined in Eqs.(5) – (7):

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\widehat{y}_i - y_i\right)^2} \tag{5}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\frac{\widehat{y}_i - y_i}{y_i} \times 100 \tag{6}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(\widehat{y}_i - y_i\right)^2}{\sum_{i=1}^{n}\left(\overline{y} - y_i\right)^2} \tag{7}$$

where n is the number of predicted values, y_i is the original measured value, $\widehat{y}_i$ is the predicted value, $\overline{y}$ is the average value of the original measured values.

Besides, the concept of improved synthesis index (*ISI*) [41] is introduced as an overall performance evaluation for more accurate evaluation of the *RMSE*, *MAPE,* and R^2.

$$ISI = \frac{1}{\mathrm{n}+\mathrm{m}}\left(\sum_{i=1}^{n}\left(\frac{P_i - P_{min,i}}{P_{max,i} - P_{min,i}}\right) + \sum_{j=1}^{m}\left(\frac{P_{max,j} - P_j}{P_{max,j} - P_{min,j}}\right)\right) \tag{8}$$

where n is the number of cost types of performance measurement, m is the number of profit types of performance measurement, and P_i and P_j represent the i and j values of performance measurement. The smaller the ISI value, the higher the performance of the model.

3. Digital twin for TSHD construction safety

The dredging process of the trailing suction ship is affected by the characteristics of the construction environment such as soil quality, hydrometeorology, etc. The complexity of these characteristics results in a risky and unstable dredging process. Digital twinning technology provide a new method for dynamic monitoring and health diagnosis. The physical and virtual sensors, as well as information transmission, are integrated into the TSHD construction safety control to realize monitoring, diagnosis, and early warning analysis of the construction status. If the target parameter data is abnormal, e.g., the data source is missing or part of the data is damaged, it can be supplemented using a data-driven perspective. Furthermore if the physical model of the problem is known, the digital model and the physical model can be integrated to evaluate the construction state of TSHD.

3.1. Stable signal for construction monitoring

Various physical sensors are installed on the dredger to monitor the construction process of TSHD. However, due to the unstable construction environment, there might be measurement deviations in the physical sensors. There is also a risk of instrument failure that may result in invalid monitoring data. Such data affects the stable monitoring of construction status and causes potential safety hazards. Besides, there are still some important construction parameters that are either difficult to measure directly, or their measurement costs are high. Currently, the γ -ray concentration meter widely is used in dredging ships which has potential radiation hazards and that causes safety and environmental risks. Such techniques cannot be also used in crowded areas. At this time, the virtual sensor based on digital twin can play an auxiliary role.

The key to digital twinning technology lies in combining the digital and physical models. The virtual sensor obtains a stable virtual signal by predicting the target parameters based on the internal relationship between the monitoring data in real-time. In cases where there is a problem with the physical sensor, (see section A in Fig. 5) a certain monitoring data might be affected with a value of 0. At the same time, other sensors operate normally and collect stable and reliable monitoring data. The target data can be therefore approximated using DTDVS by model prediction This enables continuous operation of the monitoring. and ensures the safe operation of dredgers regardless of the faulty sensor operation.

3.2. Diagnosis and early warning for mechanical failures

Various subsystems of the dredger constitute a unified system. If a mechanical failure occurs in one of the subsystems, it will affect the normal operation of other systems. Therefore, it is of great practical significance to continuously monitor the operational state of the machinery and warn the mechanical failure. Given enough operational dataset the DTDVS can obtain a reliable virtual signal by a selecting appropriate model. Comparing the data of the virtual sensor and the physical sensor, the residual error is then calculated as shown in Eq. (9):

$$\varepsilon_i = \mathrm{y}_i - \widehat{\mathrm{y}}_i \tag{9}$$

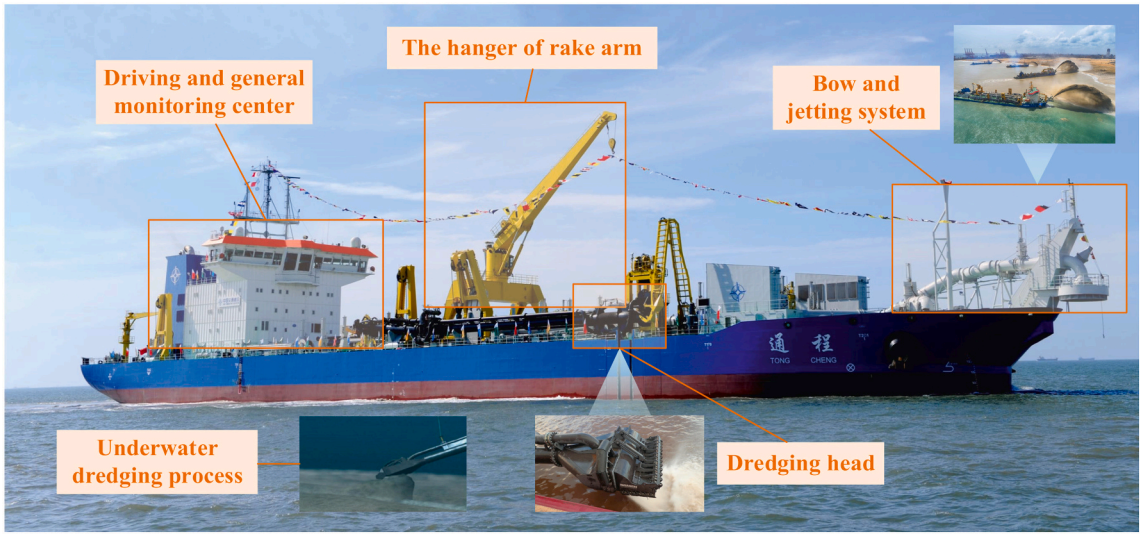


Fig. 6. "Tongcheng" trailing suction hopper dredger.

where i is the monitoring data index, y_i is the original measured value and $\widehat{y}_i$ is the predicted value.

The residual distribution under normal working conditions can be then obtained using statistical analysis. The corresponding residual warning value can be obtained by a certain degree of confidence. In the process of real-time monitoring of the TSHD construction status, the abnormal situation where the residual error exceeds the pre-warning value is treated as a warning, and the harrow operator is informed to take action accordingly.

Suppose that there is a parameter distribution family $\{F(x;\theta):\theta\in\Theta\}$, where $(X_1, X_2, \ldots, X_n)$ is a sample of the population distribution of $F(x;\theta)$. We further consider two statistics $\widehat{\theta}_i = \widehat{\theta}_i(X_1, X_2, \ldots, X_n), i = 1, 2$, defined on the sample space, Ω, and valued on the parameter space Θ. For $\widehat{\theta}_1 \le \widehat{\theta}_2$, the random interval $\left[\widehat{\theta}_1, \widehat{\theta}_2\right]$ is called an interval estimation of parameters θ. If given $\alpha(0 < \alpha < 1)$, then.

$$Pr_\theta\left\{\widehat{\theta}_1 \le \theta \le \widehat{\theta}_2\right\} = 1-\alpha, \quad \text{for any} \quad \theta\in\Theta\theta\in\Theta \tag{10}$$

where $\left(\widehat{\theta}_1, \widehat{\theta}_2\right)$ is referred to as the confidence interval in which the confidence level of θ is $1-\alpha$ and $\widehat{\theta}_1$, $\widehat{\theta}_2$ are called lower and upper confidence limits, respectively.

The occurrence of a mechanical failure can be inferred based on the value of the residual warning. For instance, in normal working conditions, the residual value of the torsional vibration is relatively small. If abnormal vibration occurs, the actual value will deviate from the predicted value and the residual error is gradually increased. At this point, it becomes necessary to issue a warning to remind the harrow operator to either adjust the ship's operating parameters or stop the machine for maintenance and inspection to avoid further deterioration of the construction failure.

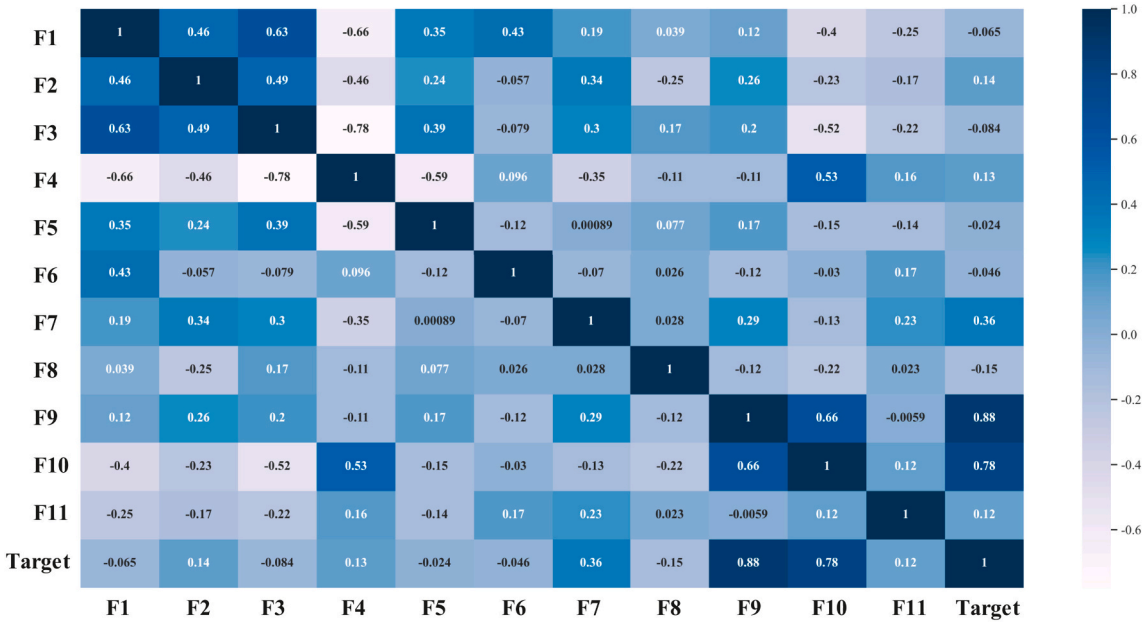


Fig. 7. Correlation analysis of the selected features.

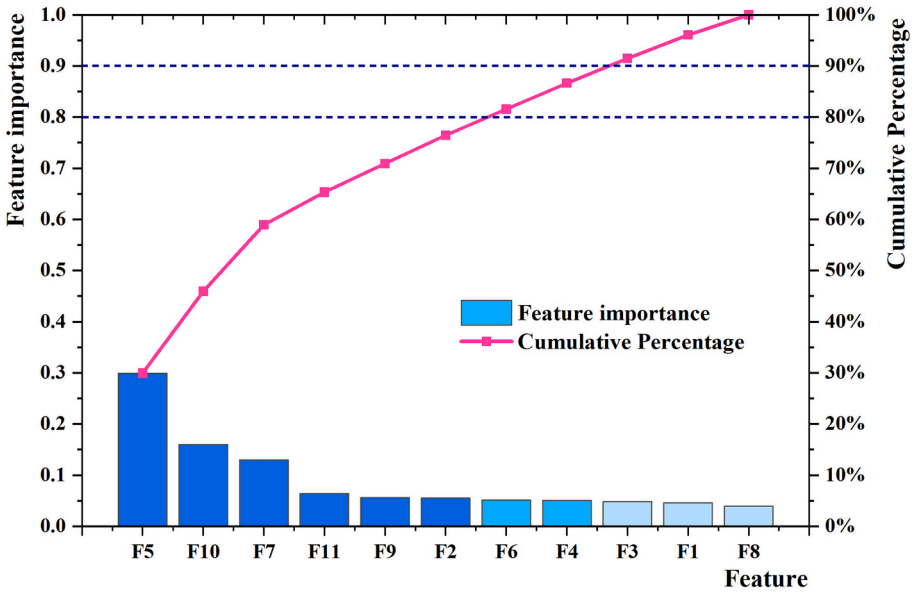


Fig. 8. Analysis of the features' importance.

4. Application and analysis

To verify the application of digital twin technology in the TSHD construction safety control, Tianjin Port 300,000-ton waterway Phase II project is considered as a case study. The channel has a total length of 44 km and a total dredging volume of 13.5 million m^3. The main soil includes muddy clay, silty clay, and sandy silt, etc. The "Tongcheng" rake suction ship (see Fig. 6), has a total installed power of 20,290 kW, a cabin capacity of 18,374 m^3, and a maximum excavation depth of 85 m. In normal operating conditions, the TSHD operates 24 h a day without interruption, and generates a set of monitoring data every two seconds. More than 100 million monitoring data records are collected within 3 days of construction operations. These data provide an important premise for machine learning models, and this paper takes them as research samples.

4.1. Virtual sensor for torsional vibration

The TSHD is often operated in complicated marine or port environments. Under the combined action of random alternating loads and current harmonics, torsional vibrations are likely to occur in mechanical equipment. Torsional vibration includes bending vibration mode and torsional vibration mode, etc. The torsional vibrations may increase the fatigue damage of bearings, reduce the service life of mechanical equipment, and produce high-level noise that harms the workers [42]; In severe cases, resonance may be created which can damage the shafting of the unit and affect the construction safety of the TSHD. Therefore, torsional vibration is an important monitoring objective of TSHD safety monitoring. Here we consider the torsional vibration as the target parameter to verify the effectiveness of the virtual sensor.

4.1.1. Feature selection and data filtering

Physical sensors provide the basis of data analysis for digital twin

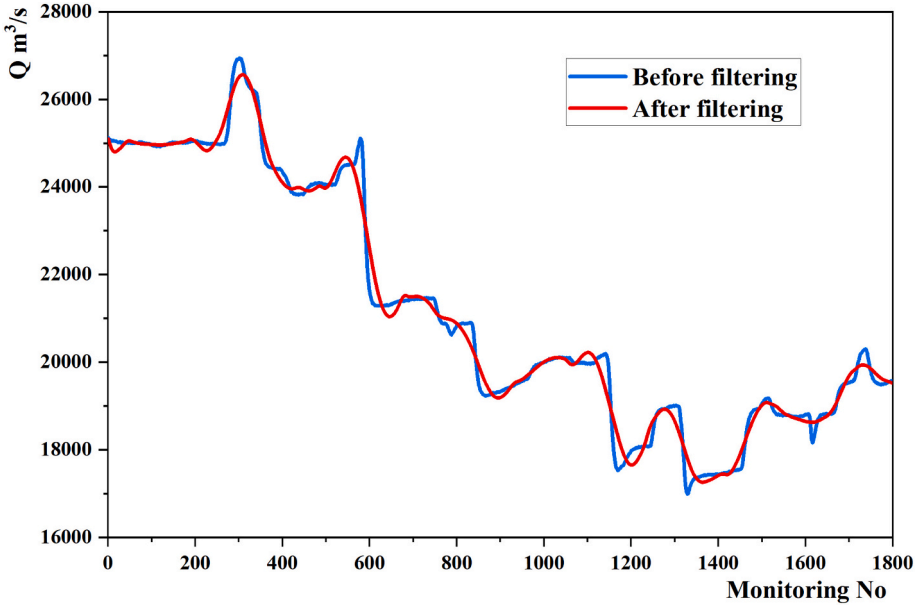


Fig. 9. Filtering effect of the flow.

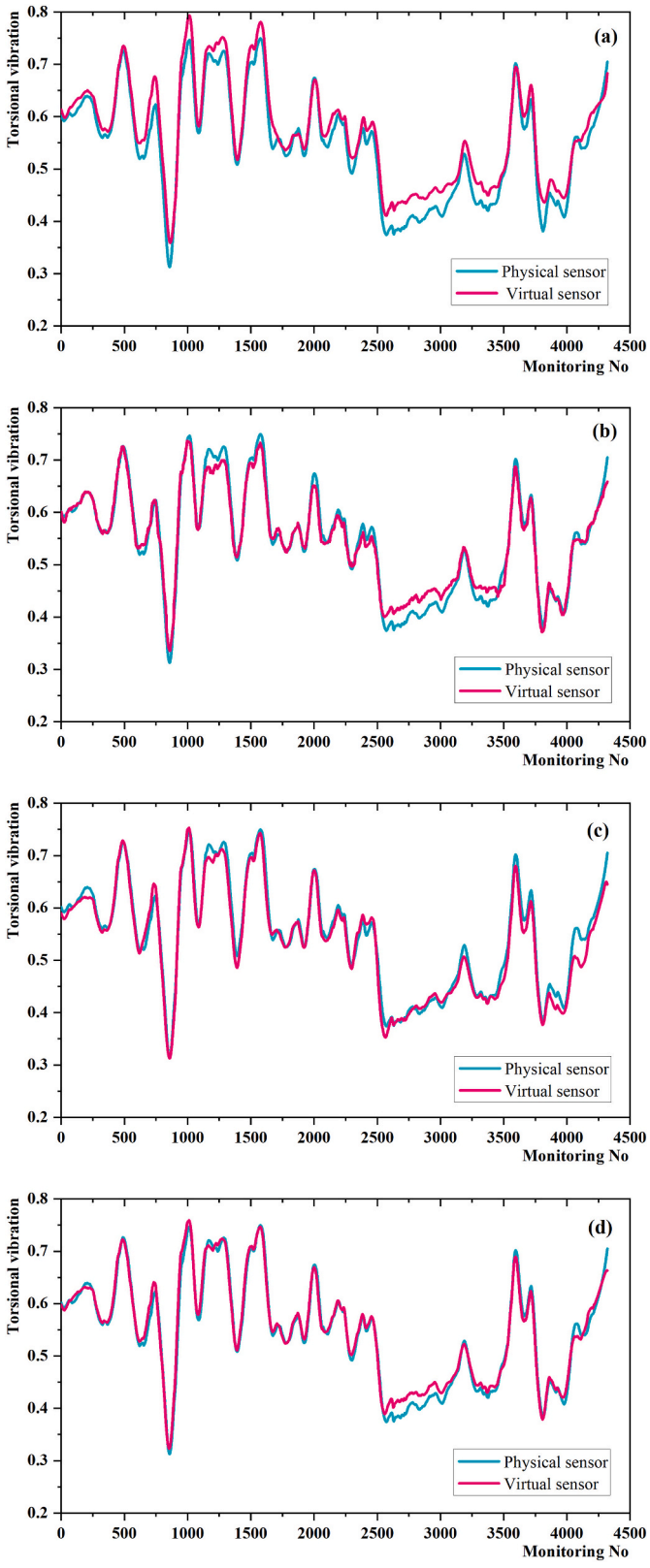


Fig. 10. Comparison between the digital and physical models: (a) prediction results of the LSTM; (b) prediction results of the XGBoost; (c) prediction results of the SVR; and (d) prediction results of the DBN.

Table 2
Performance evaluation indexes of the considered models.

Model	Evaluation index			
	RMSE	*MAPE*	R^2	*ISI*
LSTM	0.028	0.048	0.924	1
XGBoost	0.016	0.026	0.974	0.233
SVR	0.016	0.023	0.975	0.193
DBN	**0.012**	**0.019**	**0.987**	**0**

*Highlighted in bold means the best performance measure of the corresponding data mining model.

technology. The monitoring data of the construction process of the "Tongcheng" trailing suction dredger is 793-dimensional. Direct use of such high-dimensional data results in inefficient machine learning and waste computational resources. To address this issue, we use the Lasso algorithm to filter the features and obtain a smaller feature set including 27 features. Nevertheless, there is a high correlation between some of the features (some as high as 0.98). Therefore, we employ a secondary screening based on the physical meaning of the features. For instance, there is a one-to-one linear relationship between loading and displacement both of which reflect the change in the ship's weight and constitutes a correlation of 1, resulting in high feature redundancy. The optimal feature set can be then obtained by removing features with a correlation above 0.9. The heatmap of the features is given in Fig. 7.

To further analyze the influence of the feature set on torsional vibration parameters, the XGBoost algorithm is used to analyze the importance of the features. Fig. 9 illustrates the results of the importance analysis. The left ordinate represents the importance score, the right ordinate represents the percentage score, and the analysis line represents the cumulative percentage score. As it is seen in Fig. 6, the first six features are the most important, and the comprehensive contribution rate is about 80%. Among these features, "left host power" is more important and is the key performance feature. Since torsional vibration is caused by the imbalance of the torsional angle of each shaft section during the transmission of power from shafting to propeller, the related power plays a key role. Comparing Figs. 7 and 8, it is seen that a higher correlation coefficient with the target variable does not necessarily result in a greater contribution. For example, the correlation coefficient between "left flow" and "torsional vibration" is as high as 0.63, but the feature of "left flow" is only a general feature, which has little impact on the model. Therefore, more attention should be paid to the main characteristic data in the model prediction and actual fault monitoring.

We further use the Savitzky-Golay filter to filter the selected features which can reduce the noise caused by the unstable surrounding environment and measuring instruments. In Fig. 9 we present the before and after filtering of the flow feature. As it is seen, filtering results in a relatively more stable curve and can better reflect the overall trend of monitoring data. This is further conducive to improving the prediction accuracy of the model.

4.1.2. Prediction performance evaluation

The model prediction is the key to realize virtual sensor technology and its prediction performance is the main factor determining the reliability of the virtual sensor. Here we adopt four machine learning algorithms to predict the torsional vibration parameters and the results are presented in Fig. 10. The indicators mentioned above are also used to evaluate the prediction efficiency of each model, see Table 2. As it is seen, the R^2 values for all of the four considered algorithms are above 0.9. This indicates that these methods are well-responded to the variation trend of the curve, i.e., the selected features are representative. The R^2 value of the XGBoost, DBN, and SVR models are all above 0.97, however, the LSTM model has a lower prediction efficiency. In terms of the comprehensive index ISI, the DBN model achieves the highest prediction performance. Therefore, we select the DBN as the prediction model in the digital twin to realize the construction safety control of the

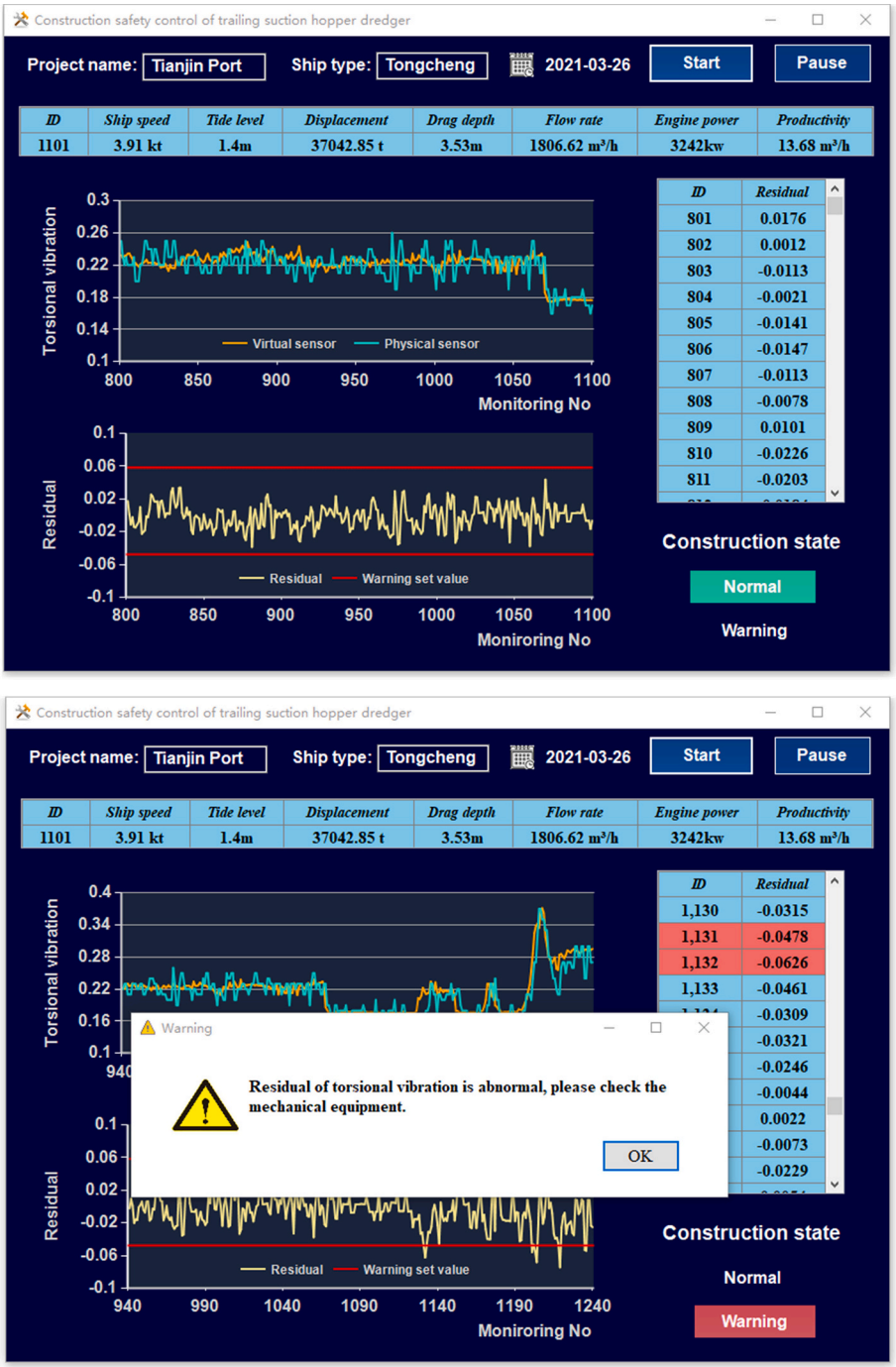


Fig. 11. The construction safety control system of the TSHD: (a) Real-time monitoring of the construction status; (b) Early warning of the abnormal situations.

TSHD.

We further analyze the statistical characteristics of the TSHD construction in normal working conditions. The torsional vibration data with the size of 8640 can be obtained by the virtual and physical sensor. The statistical results are shown in Fig. 12, where the confidence interval at 99% confidence level is [−0.0477, 0.0584]. For subsequent TSHD construction safety control, one can simply check whether the residual value is within the confidence interval.

4.2. Monitoring, diagnosing, and pre-warning system

Here we consider part of the construction stage data of Tianjin Port for analysis. We also dynamically present some of the key parameters obtained by physical sensors in the tabular form. Digital twin technology is then used to monitor the torsional vibration, an important type of mechanical failure. Both the physical sensor and the virtual sensor monitor signals and their residuals are displayed in the system as also shown in Fig. 12. In cases where there is an abnormal situation, the system sets an alarm warning the operator to take relevant measures to improve the accuracy of the construction control.

The digital twin system provides a new way for TSHD fault detection. The combination of physical sensor and virtual sensor makes fault discrimination is highly effective. The running state of the system is displayed as “Normal” (see, Fig. 11(a)). With the progress of construction, a slight torsional vibration occurs in the machinery which results in a gradual increase in the residual value in some instants it exceeds the

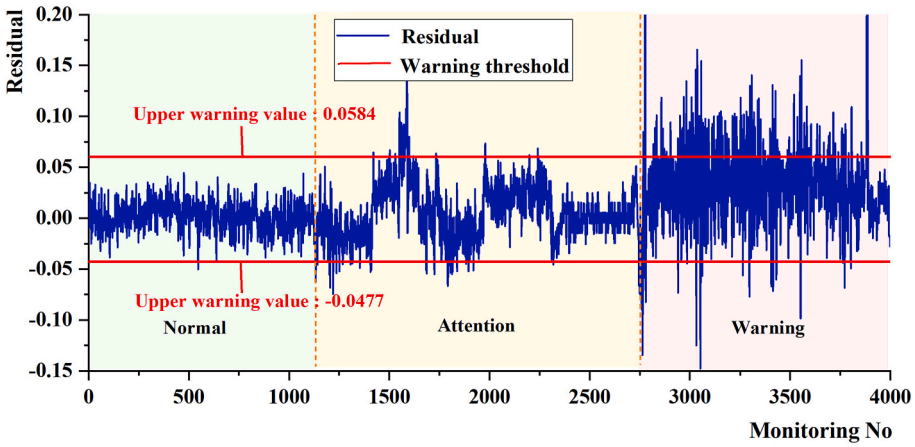


Fig. 12. Real-time monitoring of the residual signal between the physical and virtual sensors.

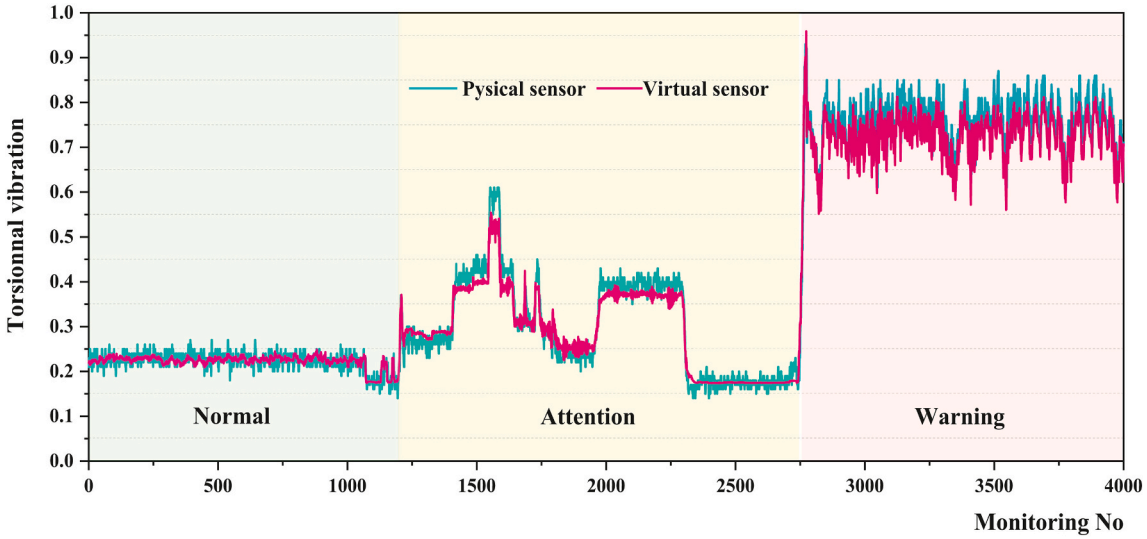


Fig. 13. Real-time monitoring of torsional vibration data using the digital twin technology.

boundary warning line. The running state is then changed to the "Warning" status as shown in Fig. 11(b). Fig. 12 shows the data trend of torsional vibration residuals which suggests that the dredger is initially in a normal operation status and the residual value is within the safety range. As it is seen, even most of the residual values remain within the warning line and the super alert point is not large. In this case, the operator is alarmed to carry out relevant inspections. The corresponding torsional vibration changes are shown in Fig. 13. Comparing torsional vibration and the residual data, it is also seen that the overall change trend of the two is consistently increasing. It also indicates that the

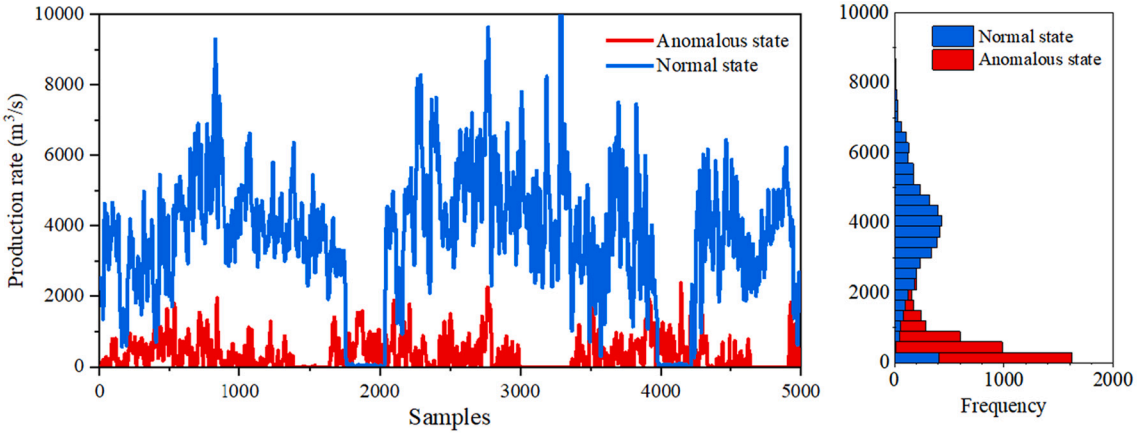


Fig. 14. The productivity before and after the accident.

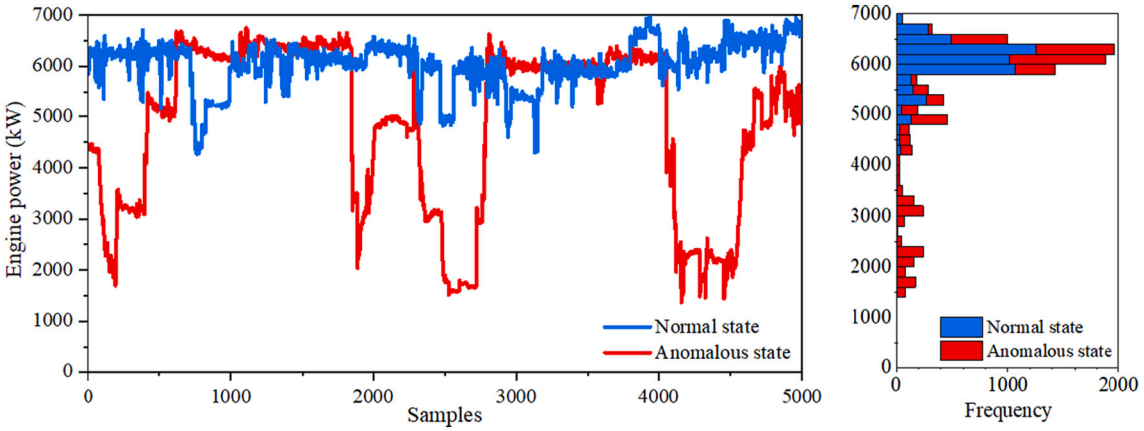


Fig. 15. The engine power before and after the accident.

method proposed in this paper can efficiently determine the abnormal point of torsional vibration. With the continuous operation of the dredger, the torsional vibration deteriorates and several residual data points exceed the critical threshold. This suggests system shutdown and full inspection recognize the fault and carry out the corresponding maintenance routine.

4.3. Discussion of the application benefits

Using the digital twin technology facilitates avoiding the risk of adopting a single physical model in construction safety control. The DTDVS proposed in this paper assists the physical sensors to dynamically monitor the construction status of dredgers and to obtain stable virtual signals of monitoring targets using a machine learning model. The combination of physical and virtual sensors enables early warning of potential mechanical faults and effective avoidance of further accidental deterioration. Without using the digital twin technology, the mechanical failure may be extended causing unnecessary economic costs.

The monitoring data of the dredger construction before and after the accident are also compared and analyzed, see Fig. 14. As it is seen, productivity, as an important index to measure the dredging benefit is significantly affected. Fig. 14 also presents the comparison of mud concentration between the later stage of failure and normal working conditions. The average value is decreased from 4025.45 m^3/s to 472.44 m^3/s, which verifies the rationality of the fault detection. As one of the main features of the model prediction, the power of the main engine also demonstrates instability after the accident as in Fig. 15. The extreme range of the data is increased from 2799 kW to 5381 kW, indicating large fluctuations. This is also consistent with the feature importance analysis presented in Section 3.1. The above results confirm that the application of digital twin technology in TSHD construction safety control effectively improves the accuracy of construction safety control.

5. Conclusions

Based on the real-time monitoring data of trailing suction hopper dredger, this paper proposed a framework of a digital twin-driven virtual sensor. In the proposed framework, the physical model was combined with artificial intelligence. Taking advantage of the two-way data interaction of digital twins, a construction safety control system was also developed for the TSHD mechanical fault monitoring system, realizing the real-time diagnosis and early warning of mechanical faults. The contributions of this paper are listed as the following:

(1) The construction monitoring data obtained by the TSHD physical sensors was analyzed and it was found that the TSHD construction data is high dimensional, noisy, and with a strong time correlation. Data preprocessing can effectively solve the above problems, which improves the prediction accuracy and efficiency of the model in the following virtual sensors.
(2) A framework of the digital twin-driven virtual sensor was then proposed. The favorable performance of DTDVS can be determined by the high prediction accuracy ($R^2 > 0.9$) of four machine learning algorithms on torsional vibration. The best performing DBN model was considered as the preferred model for the DTDVS prediction.
(3) The application of DTDVS in TSHD construction safety control was then examined. The DTDVS assists the physical sensors to monitor the target which is either difficult or costly to be directly measured. The combination of physical and virtual sensors further provides stable and reliable signals in the monitoring system, which collectively improve the accuracy of the construction safety control.
(4) We further developed a TSHD construction monitoring, diagnosis, and early warning system according to the interactive characteristics of physical and virtual sensors based on digital twin technology. Taking torsional vibration types in mechanical faults as an example, real-time dynamic monitoring was carried out. The system successfully diagnosed the abnormal states and the early warning of a mechanical fault.

Stable and reliable monitoring of TSHD construction status is an important task to improve the dredging efficiency and to ensure construction safety. Digital twinning technology overcomes the limitation of the traditional measurement method which is at the risk of abnormal functioning of the physical sensor. The developed DTDVS was shown to be more stable, environmentally friendly, and efficient in enabling the safety of dredging operations. The engineering examples also confirmed DTDVS efficiency in determining the abnormal operation of machinery and providing early warnings, thus further issuing warnings to the operator to adjust the operating parameters of the dredger or carry out the related maintenance works. Further discussions on the potential internal relations and interaction mechanisms among different types of faults might be considered as the follow-up work.

Declaration of competing interest

The authors declared that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Acknowledgments

This research was supported by the Tianjin Natural Science Foundation for Distinguished Young Scientists of China (Grant no. 17JCJQJC44000) and the National Natural Science Foundation of China (Grant no. 52179139).

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